1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist highlights that TP53 mutations in CIS create a permissive context for EMT, but not all CIS lesions progress—what molecular and microenvironmental changes (specifically E-cadherin loss and changes in stromal composition) determine which TP53-mutant CIS lesions rapidly become invasive? 2) Tier 1 paths used: I built on (a) CIS → TP53 mutations (anchors early genetic hit) and (b) EMT ↔ TP53 association plus (c) E-cadherin prognostic linkage; these directly connect an early TP53 event to EMT-mediated loss of epithelial adhesion (CDH1) and clinically measurable invasion. 3) Tier 2 support: Radiologic phenotype (ground glass nodules) and cohort-level evidence of TP53 mutations in bladder disease support that morphologic/imaging surrogates can reflect invasive biology; I treat these as supporting evidence that an imaging surrogate of stromal reaction can mark EMT-driven invasion. 4) Endpoint choice: I selected progression-free interval (PFI) because the core question concerns transition from non■invasive (CIS) to invasive disease (i.e., time-to-progression), making PFI the most mechanistically relevant and clinically meaningful endpoint. 5) Deliberate exclusions/decisions: I excluded exploratory/infectious paths (e.g., Klebsiella) from the core hypothesis because they are weakly supported and would dilute mechanistic focus; I also mapped the Ontologist's radiologic concept (ground glass nodules) to the available H&E-derived imaging feature (stromal-to-tumor ratio) as a practical surrogate for stromal/invasive signature in datasets lacking chest CT phenotype data. I excluded cross-tissue generalities (breast examples) from the core claim but retain them as secondary modifiers for generalizability checks?

Working Hypothesis

In bladder carcinoma in situ (CIS), TP53 mutation promotes epithelial-to-mesenchymal transition (EMT) with downregulation of E-cadherin (CDH1), leading to a histologic increase in stromal-to-tumor ratio (TSR) as a surrogate of a pro-invasive microenvironment. CIS tumors with TP53 mutation and low CDH1 expression, accompanied by high TSR, will have shorter progression-free interval (PFI) than CIS tumors that are TP53-wildtype or exhibit higher CDH1 and/or lower TSR.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	TP53, CDH1
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TP53: TP53 mutations in CIS create a permissive context for EMT, but not all CIS lesions progress—what molecular and microenvironmental changes (specifically E-cadherin loss and changes in stromal composition) determine which TP53-mutant CIS lesions rapidly become invasive? 2) Tier 1 paths used: I built on (a) CIS → TP53 mutations (anchors early genetic hit) and (b) EMT ↔ TP53 association plus (c) E-cadherin prognostic linkage; these directly connect an early TP53 event to EMT-mediated loss of epithelial adhesion (CDH1) and clinically measurable invasion.

CDH1: CDH1) and clinically measurable invasion.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	CIS	Supporting modifier from knowledge graph
Secondary	EMT	Supporting modifier from knowledge graph
Secondary	ground_glass_nodules	Supporting modifier from knowledge graph
Secondary	invasive_cancer	Supporting modifier from knowledge graph
Secondary	bladder_disease	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: Retrospective analysis of bladder CIS cohorts with available TP53 mutation data and CDH1 expression (IHC or RNA-based) and digital pathology to quantify TSR from standard histology slides;

Prediction 2: Cox proportional hazards models to assess the interaction between TP53 status, CDH1 expression, and TSR on PFI, adjusting for key covariates;

Prediction 3: Sensitivity analyses stratifying by TP53 mutation type and CDH1 expression levels;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- CIS: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- EMT: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.